CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

December 2011

CL 301 Transportation Engineering-I

Date: 12.12.2011, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) State & discuss briefly the basic requirements of an ideal highway alignment. **[02]**

(b) Explain Macadam's method of road construction with neat sketch. Why this method is preferred compared to other methods? **[05]**

Q - 2 (a) State different factors controlling the highway alignment. Also draw neat sketch of obligatory points controlling the highway alignment. **[02]**

(b) Explain total reaction time of driver and the factors on which it depends. Also explain PIEV theory with neat sketch. **[06]**

(c) The speed of overtaking and overtaken vehicles are 80 and 40 kmph, respectively on a two way traffic road. If the acceleration of overtaking vehicle is 1.0 m/sec^2, calculate ,Safe overtaking sight distance, Minimum length of overtaking zone andDesirable length of overtaking zone. Also draw a neat sketch of the overtaking zone and show the positions of the sign posts. **[06]**

OR

Q - 2 (a) Describe in brief the steps to be adopted in a new highway project work. **[02]**

(b) Define Super elevation. Derive the expression $e+f = v^2/gR$ assuming suitable data. **[06]**

(c) A horizontal circular curve having radius 325m, design speed = 65kmph, length of wheel base of largest truck = 6m and pavement width = 10.5m, Design Super-elevation, Extra Widening of Pavement andLength of Transition curve. **[06]**

Q - 3 (a) Enlist different tests for Bitumen. **[02]**

(b) Write a detailed note on Road User characteristics. **[06]**

(c) List out different traffic studies to be carried out in a highway project. Explain in detail Traffic Volume Study, its objectives and uses. **[06]**

OR

Q - 3 (a) Enlist the tests for Road Aggregates. **[02]**

(b) Write a note on Origin & Destination Studies. Also discuss various applications of **[06]**

O-D studies.

(c) Write a detail note on Traffic Signs. Draw at least 5 diagrams of each type of Traffic Signs. **[06]**

SECTION – II

Q - 4 (a) Write a note on California Bearing Ratio (CBR) Test (Neat sketch required). **[04]**

(b) Define any three from the following: (Draw sketch wherever needed) **[03]**

1. Prime Coat
2. Water Bound Macadam
3. Built Up Spray Grout
4. Penetration Macadam

Q - 5 (a) Write a short note on construction of WBM roads. **[03]**

(b) Design Flexible Pavement as per IRC guidelines for a new road considering the following data.Two lane single carriageway, Initial Traffic in the year of completion of construction = 400 CVPD, Traffic Growth rate per annum = 8 %, Design Life = 15 years, Vehicle Damage Factor = 4, Design CBR of Sub grade Soil = 5 %. **[11]**

OR

Q - 5 (a) Write a note on construction of Bituminous Roads. **[03]**

(b) A Cement Concrete pavement is to be designed as per IRC guidelines for a two lane two way NH in Gujarat State. The total two way traffic is 3000 CVPD at the end of construction period. Use Flexural Strength of Cement Concrete = 45 kg/cm^2, Effective Modulus of sub grade reaction of DLC sub-base = 8 kg/cm^3. Elastic Modulus of concrete = 3 x 10^5 kg/cm^2, Poisson's Ratio = 0.15, Coefficient of Thermal Expansion of concrete = 10 x 10^{-6} /^{0}C, Tyre pressure = 8 kg/cm^2, Rate of increase of Traffic = 8 %, Spacing of Contraction joints = 4.5 m, Width of Slab = 3.5 m. **[11]**

The axle load spectrum obtained from axle load survey is given as follows:

Single Axle Loads		Tandem Axle Loads	
Axle Load Class, Tons	% of Axle Loads	Axle Load Class, Tons	% of Axle Loads
19-21	0.6	34-38	0.3
17-19	1.5	30-34	0.3
15-17	4.8	26-30	0.6
13-15	10.8	22-26	1.8
11-13	22.0	18-22	1.5
9-11	23.3	14-18	0.5
Less than 9	30.0	Less than 14	2.0
Total	93.0	Total	7.0

Q - 6 (a) Define surface and sub-surface drainage. **[09]**

The maximum quantity of water expected in one of the open longitudinal drains on clayey soil is 0.9 m^3/sec. design the cross section and longitudinal slope of trapezoidal drain assuming the bottom width of the trapezoidal section to be 1.0m and cross slope to be 1.0 vertical to 1.5 horizontal. The allowable velocity of flow in the drain is 1.2 m/sec and Manning's roughness coefficient is 0.02.

(b) Explain sources of revenue by which funds are collected by different authorities in detail. **[05]**

OR

Q - 6 (a) The surface water from road side is drained to the longitudinal side drain across one half a bituminous pavement surface of total width 7.0 m, shoulder and adjoining land of width 8.0 m on one side of the drain. On the other side of the longitudinal drain, the water flows across from reserve land with grass and 2% cross slope towards the side drain, the width of this strip of land being 25 m. the run-off coefficients of the pavement, shoulder and reserve land with grass surface are 0.8, 0.25 and 0.35 respectively. The length of the stretch of land parallel to the road from where water is expected to flow to the side drain is about 400 m. Estimate the quantity of run-offflowing in the drain assuming 25 years period of frequency.Design the cross section and slope of the side drain in loamy soil with Manning's roughness coefficient = 0.022 and suitable speed of flow = 0.8 m/sec. **[09]**

(b) Write and explain the methods to recollect finance which was used in high Way finance **[05]**

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